

UCab — Full-Stack MERN Cab Booking Platform

Solo Developer / Team Member: Shaik Sumiya Zainab | Project ID: N/A (Solo Track)

Phase 2: Requirement Analysis — Customer Journey Map

Project Name: Cab Booking (`UCab`)
Project ID: `N/A (Solo Track Submission)`
Team Member / Solo Developer: Shaik Sumiya Zainab

1. End-to-End Rider Journey Map

The customer journey map details the step-by-step experience of a rider using UCab, from initial discovery to ride completion and receipt generation.

```
journey
title UCab End-to-End Rider Journey
section 1. Onboarding & Authentication
Visit Landing Page (`/home`): 5: User
Select 1-Click Demo Login (`/login`): 5: User
Receive Welcome Toast & Session Sync: 5: User
section 2. Discovery & Selection
Explore Vehicle Inventory (`/cabs`): 5: User
Sort by Price & Category (Sedan, SUV): 5: User
Select Preferred Cab & Click Book: 5: User
section 3. Booking & Customization
Enter Pickup & Drop Locations (`/book/:id`): 5: User
Apply Promo Code (`UCAB20` for 20% Off): 5: User
Select Refreshments (`+ 150`) & Charity (`+ 120`): 5: User
Confirm Automatic Payment Checkout: 5: User
section 4. Real-Time Tracking & Ride
Redirect to Live GPS Progress Bar (`/home-user`): 5: User
Inspect Assigned Driver (`Vikram Sharma`): 5: User
Receive Status Updates (`Accepted` -> `Started`): 5: User
section 5. Completion & Receipts
Trip Marked Completed by Dispatch: 5: User
Visit Trip History (`/bookings`): 5: User
Download Corporate PDF Receipt with QR: 5: User
```

2. Touchpoint & Opportunity Analysis Table

Journey Stage	User Goal	Touchpoint / Component	Potential Pain Points	UCab Engineering Solution
1. Onboarding				
	Quick, frictionless login without typing lengthy credentials.	`Login.jsx` & `AuthContext.jsx`	Forgotten passwords or slow email verification during demos.	1-Click Demo Login: Instantly injects pre-seeded credentials (`sumiya@ucab.com`) and issues JWT session.
2. Discovery				
	Compare prices and vehicle capacities across categories.	`CabListing.jsx`	Overwhelming vehicle lists without clear pricing distinctions.	Filter & Sort Engine: Interactive category tabs (`Mini`, `Sedan`, `SUV`, `Luxury`) and sort by `1/km`.
3. Booking				
	Customize trip fare and request extra travel comforts.	`BookCab.jsx`	Hidden fees or lack of transparency on distance calculation.	Real-Time Fare Breakdown: Live distance calculation with instant promo code subtraction and toggleable perks.
4. Tracking				
	Know where the driver is and when the cab will arrive.	`UserHome.jsx`	Uncertainty about whether the trip was accepted or dispatched.	Live Dispatch Status Bar: Visual 4-stage tracking indicator with driver contact number and vehicle plate.
5. Post-Ride				
	Obtain verified receipts for corporate tax expense filing.	`MyBookings.jsx` & `ReceiptModal.jsx`	Clunky email invoicing or missing tax breakdown details.	Corporate QR Invoice Modal: One-click printable thermal receipt with embedded verification QR code.